



Dr. Jacob Freeman
Associate Professor
Anthropology Program
Utah State University
0730 Old Main Hill
Logan, UT 84332

Promotion Committee
Department of Sociology, Anthropology, & Criminal Justice
Utah State University

October 15, 2024

Dear Reviewers,

I am writing to ask for your recommendation for promotion to the rank of Full Professor. I am currently an Associate Professor in the Department of Sociology, Anthropology, and Criminal Justice at Utah State University (USU). My role statement is 50 % research, 40 % teaching, and 10 % service. As a researcher, I build and evaluate models, within ecological and evolutionary frameworks, to study the long-term drivers of human population growth and why some groups govern shared resources more effectively than others in the face of stress. To date, I have published 46 peer reviewed papers (17 as Associate Professor) and led or co-led six competitive research grants (two as Associate Professor) totaling \$2,316,742 in funding. As an undergraduate educator, I motivate students to think critically about how the interaction between human institutions, individual decision making, and ecosystems affects the capability of human groups to cope with environmental change. As a graduate educator, I help students prepare for a professional career in applied archaeology that is informed by anthropological theory. As an academic citizen, I serve as Program Director for the Anthropology Program at USU (expected term 2020-2025). I also serve as an Academic Editor for *PloS One*, peer review papers, organize sessions at professional conferences, and serve on university and professional committees.

I believe that USU is an excellent university, and I hope to continue my career at USU as a Full Professor in the academic community. Please find appended to this letter a summary of my research, teaching, and service activities; my *curriculum vitae*; and a sample of published papers since my appointment as an Associate Professor. Thank you for your consideration.

Sincerely,

Jacob Freeman

Research Summary

As an Associate Professor, I have established a research program with accelerating impact and an international reputation for studying two related and important questions. (1) Why do human populations grow and decline over space and time? And (2) why do some social groups collectively govern shared resources effectively in stressful situations while others do not? Answering these questions helps explain how human populations respond to social-ecological challenges (e.g., climate change), and how the stress generated by such challenges impacts (or not) the ability of social groups to govern shared resources in complex environments. My research program has led to a robust record of granting and publication. Since my appointment, I have led or co-led six competitive research grants (two as Associate Professor) totaling \$2,316,742 in funding, and I have authored 46 peer reviewed articles (17/46 since my appointment as Associate Professor). According to rankings compiled by Scimago Journal & Country Rank (SJR), 39/45 articles are published in journals ranked in the top quartile of all journals in their respective disciplines (two in the second quartile and four in edited volumes). As an Associate Professor, I have published an average of 4.25 manuscripts per year (Figure 1a). To date, according to Google Scholar, my publications have accumulated 940 citations, with an H-index of 19, and an i10-index of 35. Since my appointment, my peer-reviewed publications have increased linearly, and citations of this work continue to increase geometrically (Figure 1B).

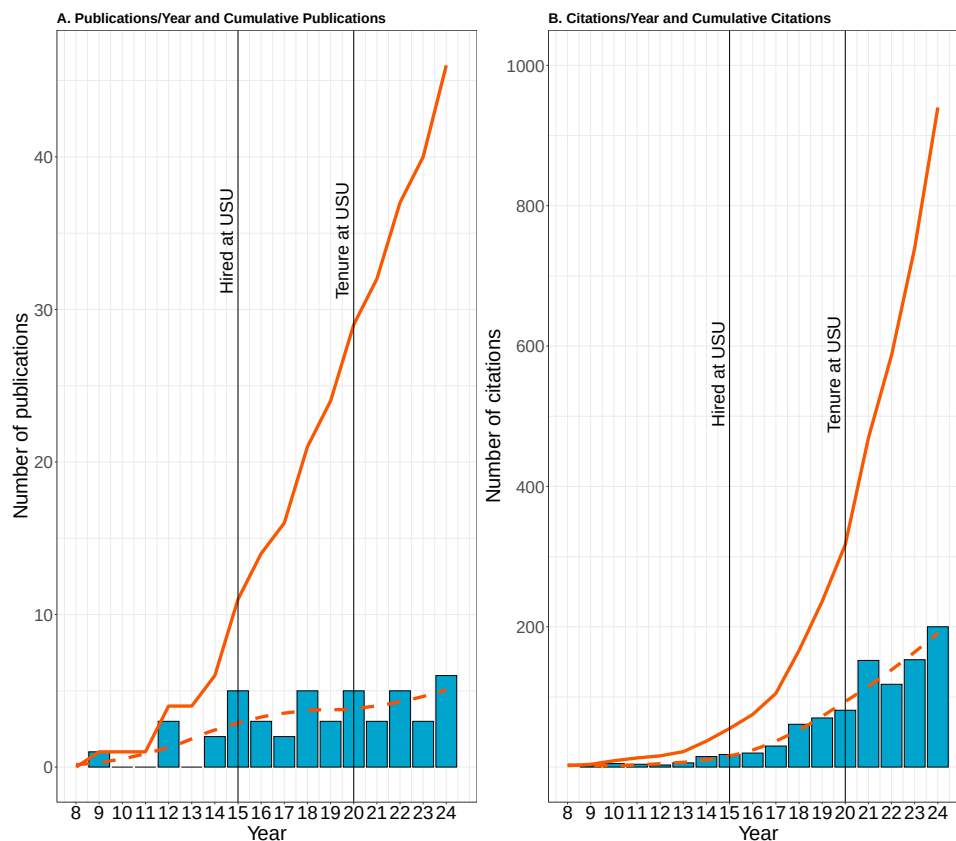


Figure 1: Research output: (a)-publications per year (blue bars) with loess trend (dashed orange line) and cumulative publications (orange line); (b) citations per year (blue bars) with loess trend (dashed orange line) and cumulative citations (range line). Note: citation and publication data retrieved from Google Scholar October 15, 2024.

Research program

(1) I lead a global network of archaeologists, paleoecologists, and mathematicians called PEOPLE 3000. The impetus for this group was a realization that our understanding of human population ecology was severely limited by a dearth of data on long-term changes in human population and proper tools for interpreting those data. Thus, as an Assistant Professor, I began organizing colleagues and publishing papers that established a vision for a comparative population ecology of human societies over the long-term. The first paper was published in *PNAS* and compares changes in human population across four continents over the last 10,000 years. The paper illustrates that human populations boom and bust in unison on the same continents, suggesting a great antiquity to the process of mutual dependence among societies.

As an Associate Professor, I have built on this foundation, publishing 12 papers that contribute to the theory and data necessary to help explain the long-term growth and decline of human populations. For example, as an Associate Professor I co-led (please see Bird et al., 2022, on CV) and contributed to (please see Kelly et al., 2022, on CV) the publication of large data sets of archaeological radiocarbon useful for reconstructing human population dynamics over space and time. The development and publication of these datasets is a major contribution shaping the questions that archaeologists can ask about human population ecology and the ability of archaeologists to propose and test models of long-term population growth. For instance, I recently led a paper published in *PNAS* that uses the above datasets to study the long-term expansion and recession of human populations in North America. The paper builds on a series of publications during my appointment as an Associate Professor (e.g., Freeman et al., 2023b, 2024b; Peralta et al., 2023, 2022; Freeman et al., 2021b,a, on CV) and provides a framework for evaluating the factors that impact the long-term growth and stability of human populations across archaeological regions. We find that innovations in technology and institutions drive long-term population expansion among hunter-gatherers and agricultural societies; however, agricultural regions experience more consistent innovation, population expansion, and, ultimately, more severe population recessions than hunter-gatherer regions. These results provide a foundation for future studies of the multi-scalar process of technological innovation and how innovations interact with climate change to impact human populations (please see Freeman et al., 2024a, on CV).

(2) I lead the Cognitive Styles and Collective Action project. This project studies how feedbacks between environmental change, cognition, and biological responses to stress impact the ability of groups to successfully govern shared resources. As an Associate Professor, I have published four papers as a part of this project. These studies illustrate the effects of systematizing and empathizing—two distinct information processing abilities—on the ability of groups to govern shared resources. Groups with *high levels of both systematizing and empathizing* govern resources more effectively when confronted with unexpected, negative changes in their environment than groups with *a highly specialized ability in systematizing or empathizing*. My continuing work studies basic feedbacks between social institutions, cognition, and biological resources. For instance, in a recent paper we document that when societies establish institutions that lower the energetic costs of empathizing and building trust, they are more productive (e.g., higher GDP). However, this productivity comes with a cost, more connected and trusting societies also must extract more energy from their environment to sustain larger, more connected groups over time (please see Freeman et al., 2023a, on CV). Building on this work, I led the development of a funded NSF grant that studies how feedbacks between a shared resource, resource users, and

stress arousal impacts the sustainability of critical ecosystems (e.g., the Colorado River Watershed). We model how stress arousal mediates that ability of individuals to use their cognitive tools to communicate and imagine future resource scenarios, generating tipping points in the persistence or failure of collective action (please see 2024 NSF, DRMS grant and Baggio et al. 2024 on CV).

My grant activity reflects a commitment to supporting research on human responses to environmental change across multiple scales and supporting student research. The specific grants include: (1) **March 2024-February 2027, Role PI.** NSF, DRMS Collaborative Proposal: Group cognition, stress arousal, and environment feedbacks in decision making and adaptation under uncertainty. NSF DRMS Program, total of \$630,000. The success rate of grants submitted to this competition was less than 15 %. This grant investigates how stress arousal mediates that ability of individuals to use their cognitive tools to communicate and imagine future resource scenarios, generating tipping points in the persistence or failure of collective action in shared resource settings. (2) **August 2017-August 2025, Role PI.** PEOPLE 3000 Scientific Working Group sponsored by Past Global Changes (PAGES), \$35,000 (to date) in workshop support. This working group brings together a global network archaeologists, paleoecologists, and mathematicians to study the long-term population ecology of human societies (<http://www.pastglobalchanges.org/ini/wg/people-3000/intro>). (3) **Funded September 2016 Role: PI.** General Intelligence, Social Intelligence, and Robust Cooperation in Social-Ecological Systems NSF, IBSS-L, \$465,892. The success rate for grants submitted to this competition was less than 13 %. This grant proposes to assess the effects of diverse cognitive abilities on the ability of individuals to self-organize and cooperate to manage resources, even when stressed by an unexpected social or ecological change. (4) **Funded September 2015, Role: Co-PI.** The Evolution of Social Networks and the Robustness of Human Societies to Population Growth and Climate Change: A deep time perspective. NSF, IBSS Program; \$998,350. The success rate of grants submitted to this competition was less than 15 %. Through empirical analysis and mathematical modeling our team is evaluating the hypothesis that selective pressures favoring the evolution of territorial ownership and associated exchange networks minimized variance in the variance of an individual's flow of resources over the short-run, but also set forager-resource systems on the Texas Coastal Plain up for cascades of social dissolution. (5) **Funded August 2015, Role: Co-PI.** 6000 Years of Hunter-Gatherer Territoriality: Isotopic and Genetic Analysis of the Morhiss Site, Texas. NSF, BCS Archaeology Program; \$216,000. This grant seeks to understand the evolution of territorial ownership and subsistence economies in hunter-gatherer societies on the prehistoric Texas Coastal Plain.

In addition to the above grants, I am also currently working on a NSF grant that attempts to explain the deep-time biogeography of maize as a response to basic biophysical variables (elevation, temperature), long-term population growth and decline, and the social diversity of seed exchange networks. The grant builds on the synthetic model of long-term population growth (please see Freeman et al., 2024c, on CV) to help explain patterns of innovation in agriculture and maize biodiversity and the adaptability of farmers to climate change.

Finally, I have presented my research at 14 advanced seminars/invited presentations and given 32 presentations at national and international conferences. My lab has a robust participation of students, with undergraduate and graduate students leading or co-presenting 11/32 presentations. These are fun, rewarding collaborations that create synergy between my research and teaching activities.

Final research statement

Through publications, presentations, and granting, I have established an international reputation for interdisciplinary research and publishing on topics of theoretical importance. Moving forward, I would like to build on the core research projects above. Specifically, I will write a monograph that summarizes what we have learned from building the largest, globally comprehensive data set assembled the purpose of studying human population ecology. Further, I will summarize the results of ethnographic comparative material that I have collected over the last 15 years. This material lies at the foundation of the population models that I have developed over the past three years and can provide an excellent resource for archaeologists. I will also continue to write papers and grants on how cognition, stress, and environmental change interact to impact the governance of shared resources. I plan to continue collaborating within the Anthropology Program and across colleges at USU.

Teaching & Mentoring Summary

I have two goals as an educator. First, I provide students with the opportunity to participate in research and think critically about human responses to population growth and climate change. I strive to create an environment where students engage their *imaginations* to develop explanations for social-ecological patterns. The skills required to develop such explanations transfer across domains of inquiry and empower lifelong learning. For instance, in the Archaeology of Ancient Civilizations, I developed an assignment that asked students to develop their own explanation for the adoption of agriculture by hunter-gatherer societies and compare their explanations with those of professional anthropologists. I then presented my own work on this research problem. In subsequent years, based on student feedback, I redesigned this assignment. I first asked students to compare a set of ethnographic readings that describe contemporary hunter-gatherer and agricultural societies. Next, in groups the students coded the similarities and differences between the societies. Finally, based on this coding experience, they imagined why hunter-gatherers may or may not adopt agriculture. We then compared the students' explanations with those of professional anthropologists. The revision of the assignment to ask students to make observations from ethnography and analyze the differences between foragers and farmers improved student participation and satisfaction with the assignment. Almost all student explanations include discussions of human responses to climate change and population growth.

Second, I assess and adapt my lessons and materials to student learning and outcomes. This process occurs at multiple scales. At the scale of an individual class period, I ask questions and solicit written feedback to adjust class activities based on student responses. During the semester, I use exams, written and oral assignments, and surveys to assess how students are progressing on course objectives. Finally, between semesters, I assess and change course structures to better communicate with students and motivate learning. For example, I teach Histories and Theories of Anthropology every spring semester. Beginning in 2022, along with a change in course delivery, I noticed a marked decline in the quality of essays written in the class. The poor performance continued in 2023 despite changes in my class presentation and teaching. After interviewing students, I found that student time spent reading the primary literature assigned in the class decreased markedly. Thus, I redesigned the class in 2024 to focus on five key concepts in Anthropology (Evolution, Culture, Structure-Agent, Function, and Epistemology). I then used a general reader to introduce each concept, supplemented the reader with primary

literature, and asked the students to search the literature for other examples. Further, I scrapped all essays and created an assignment structure that creates a continuous dialog across the semester. When first introducing key concepts, students now write graded discussion posts that allow us to dialog both in writing and in class. Next, instead of two essays that synthesize the readings, students create a concept map that describes the relationship between the concepts discussed above and key authors. The students then present these maps in class, creating a dialog between all of us about how the concepts fit together. Building on the concept maps, students then take an exam that compares two papers from the primary literature published in the last five years. This exam asks students to identify and critique the use of the key concepts in contemporary anthropology. This course structure has better accommodated students taking the course via multiple modalities and led to better engagement with the readings and synthetic thinking that are critical objectives of the class (please see appended peer teaching evaluations).

My philosophy of graduate education focuses on providing students with opportunities to engage in research design. I believe that graduate students are apprenticing colleagues, and I seek to help them become knowledge producers through continual practice asking research questions, answering those questions through formal models and/or the collection and analysis of empirical data, and writing the results up for publication. To date, my students have coauthored 15 published articles and 11 presentations.

I seek feedback from graduate students and adjust my course designs to better help them achieve the skills needed to become productive, professional archaeologists. For example, in my quantitative methods course, I introduced students to the computing language R. This language is very useful for quantitative and spatial analyses. However, students found learning the language and quantitative concepts challenging, especially using a course format of demonstration and then practice assignment. Thus, I redesigned the course around three research projects. I introduced R. Then I introduced the first research project. This project required students to build graphs in R (exploratory data analysis) and assess if the data were consistent with a hypothesis discussed in class. Next, I introduced a second research project, building on the first. This required students to, again, conduct an exploratory data analysis, plus use more advanced quantitative tools to assess if the data were consistent with a hypothesis discussed in class. Finally, the third project reinforced skills developed in the first two projects and required additional analysis. This redesign gave the students repeated exposure to both the computing language and key concepts.

As an Associate Professor, I have integrated the use of R into my other graduate courses to reinforce skills developed in the quantitative methods class across the curriculum. I have observed a marked increase in proficiency in the use of R among the students as well as more developed quantitative reasoning. For example, in 2022, I used a research based course design in my Paleodemography graduate course. In this course, we focused on two projects (1) building a continent wide data set of human bone stable isotopes to study the ecological determinants of sex differences on diet and (2) building a data base of resource use among ethnographically recorded societies to study the potential effect of population density on human diet breadth. Both projects provided the students with the opportunity to learn key concepts and models in demography, collect data, apply models to analyze data, and present their findings at the Society for American Archaeology meetings.

In sumary, I teach graduate and undergraduate courses at USU and mentor undergraduate and

Table 1: Idea scores and ratings

	Progress on Objectives	Excellent Teacher	Excellent Course	Summary Evaluation	Comparison Group
Mean IDEA Score	56	53	52	55	IDEA Database
Comparison to Other Faculty	Higher	Similar	Similar	Similar	IDEA Database
Mean IDEA Score	54	53	51	53	Discipline
Comparison to Other Faculty	Similar	Similar	Similar	Similar	Discipline

graduate students. I have taught 25 courses at USU and mentored 15 graduate students to the completion of their degree. Table 1 displays my IDEA score evaluations against two comparison groups (all faculty in the IDEA database and anthropology only). The scores indicate that I am a proficient teacher, similarly rated to other teachers in the IDEA database. In addition to teaching, I have served on the completed Masters Thesis Committees of 15 students, and as chair of four Masters Thesis Committees, and one completed PhD committee. As an undergraduate mentor, I have chaired four formal summer research grants. Samantha Nabity won a College of Humanities and Social Sciences Summer Research Grant from USU. I mentored her through the project, and she presented the results at the Society for American Archaeology meetings in 2019. Her experiences as a part of this fellowship helped her earn a high paying position in analytics for an insurance company. I also mentored Lux Miranda through a Peak Summer Fellowship (a \$4,000 award). The outcome of Lux's fellowship was their first publication—in *PloS One* and provided the foundation for their Masters degree and current pursuit of a PhD in social robotics. I currently mentor four undergraduate research projects. For example, Matthew Jensen presented their findings on the effect of inequality on population collapse at the Society of American Archaeology meetings in 2024, and Matthew is continuing this work with Neil Dstrop to study the relationship between storage, population overshoot, and landscape modification in the American Southwest. They will present their results at the Society for American Archaeology meetings in 2025. Alyssa Ahmann, winner of the inaugural Simms Undergraduate Research Scholarship (\$2,500), is studying sex differences in protein consumption during periods of lower and higher population density among hunter-gatherers and farmers. The initial results of her work will be submitted for publication in late 2024 and presented at the Society for American Archaeology meetings in 2025.

It is a privilege to teach courses, mentor students, and witness their success. Moving forward, I will continue to engage in reflexive teaching in an effort to help students develop critical thinking skills and build life-long learning habits.

Service & Outreach Summary

Since the fall of 2020, I have served as Program Director for the Anthropology Program at USU. As Program Director, I have led the development and execution of the Anthropology Program's mission through curriculum development, scheduling, promotion of the program through events, advertising, and promoting a sense of community. Major accomplishments include:

- Submitting a R401 for a PhD and Graduate Certificates in Applied Archaeology and PhD program. The PhD and Graduate Certificates have been approved and begin accepting students in the Fall of 2024.

- Restarting the Anthropology Celebration in 2021. Since the celebration began, we have held five events that have averaged over 40 participants, including students, faculty, and alumni.
- Redesigning our office advertisement materials and display space and instituting the first alumni newsletter and survey. The newsletter is published once per semester and helps connect alumni with the program.
- Working with donors to revise and establish the Simms Undergraduate Research Scholarship. This scholarship provides an anthropology student with \$2,500 to conduct undergraduate research each year.

On going work as Program Director includes: Revising the Anthropology website, revising the program's 5-year plan, and putting together an archaeological field school program. In addition to these activities, I also manage the academic schedule for faculty, transfer credit articulations for students, and manage approvals for undergraduate student requests for exceptions.

Since my appointment, I have served as a peer reviewer for more than 48 journal articles, as well as for the Swiss and US National Science Foundations. I have also written one book review and one commentary on an article for *Current Anthropology*. Since 2020, I have served as an Academic Editor for *PloS One*. As editor, I evaluate approximately two manuscripts per month. I have also served on 15 college and department level committees. Currently, I chair one promotion and tenure committee, and I sit on four additional promotion and tenure committees as a regular member. As part of this service, I write teaching evaluations for tenure candidates and provide feedback on their promotion and tenure materials. I also serve as an Ombudsperson for tenure and promotion committees at USU, sitting in on 2-4 committee meeting per year.

In addition to my service to the academy and to Utah State University, I am also engaged in public outreach. Since my appointment, I have conducted public lectures at the Utah State Museum of Anthropology, The Leonardo Museum in Salt Lake City, Utah, and Science On-Tap in Logan, Utah. I have also met with an USU alum to discuss the potential to donate a collection of ceramics to the USU Museum of Anthropology. Finally, I have given several interviews on Utah Public Radio, including on the programs Access Utah.<http://upr.org/post/iq-and-social-intelligence-wednesdays-access-utah> and Undisciplined, and I have also given interviews on my research with the local news radio station KVNU.

Concluding Statement

My current appointment as an Associate Professor at Utah State University is a privilege. The position provides the opportunity to live the values of intellectual curiosity and enthusiasm for learning. The thrill of learning and the importance of imagination and analysis to gain new insights are key values that I attempt to motivate in undergraduate and graduate students alike. Through editing, reviewing articles, and serving on university committees, I hope to motivate these values among friends and colleagues as well. Thank you for reviewing my portfolio.

References

- Bird, D., Miranda, L., Vander Linden, M., Robinson, E., Bocinsky, R. K., Nicholson, C., Capriles, J. M., Finley, J. B., Gayo, E. M., Gil, A., Freeman, J., et al. (2022). p3k14c, a synthetic global database of archaeological radiocarbon dates. *Scientific Data*, 9(1), 1–19.
- Freeman, J., Anderies, J. M., Beckman, N. G., Robinson, E., Baggio, J. A., Bird, D., Nicholson, C., Finley, J. B., Capriles, J. M., Gil, A. F., Byers, D., Gayo, E., & Latorre, C. (2021a). Landscape engineering impacts the long-term stability of agricultural populations. *Human Ecology*, 49(4), 369–382.
- Freeman, J., Baggio, J. A., Miranda, L., & Anderies, J. M. (2023a). Infrastructure and the energy use of human polities. *Cross-Cultural Research*, 57(2-3), 294–322.
- Freeman, J., Gil, A. F., Peralta, E. A., Franchetti, F., López, J. M., & Neme, G. (2024a). A model of long-term population growth with an application to central west argentina. *PloS one*, 19(8), e0307703.
- Freeman, J., Hard, R. J., Mauldin, R. P., & Anderies, J. M. (2021b). Radiocarbon data may support a Malthus-Boserup model of hunter-gatherer population expansion. *Journal of Anthropological Archaeology*, 63, 101321.
- Freeman, J., Mauldin, R. P., Hard, R. J., Solis, K., Whisenhunt, M., & Anderies, J. M. (2024b). Hunter-gatherer population expansion and intensification: Malthusian and boserupian dynamics. *Journal of Archaeological Method and Theory*, 31(3), 761–781.
- Freeman, J., Mauldin, R. P., Whisenhunt, M., Hard, R. J., & Anderies, J. M. (2023b). Repeated long-term population growth overshoots and recessions among hunter-gatherers. *The Holocene*, 33(10), 1163–1175.
- Freeman, J., Robinson, E., Bird, D., Hard, R. J., Mauldin, R. P., & Anderies, J. M. (2024c). The long-term expansion and recession of human populations. *Proceedings of the National Academy of Sciences*, 121(12), e2312207121.
- Kelly, R. L., Mackie, M. E., Robinson, E., Meyer, J., Berry, M., Boulanger, M., Coddling, B. F., Freeman, J., Garland, C. J., Gingerich, J., et al. (2022). A new radiocarbon database for the lower 48 states. *American Antiquity*, 87(3), 581–590.
- Peralta, E. A., Freeman, J., & Gil, A. F. (2023). Population expansion and intensification from a malthus-boserup perspective: A multiproxy approach in central western argentina. *Quaternary International*.
- Peralta, E. A., López, J. M., Freeman, J., Abbona, C., Franchetti, F., Ots, M. J., Cahiza, P., Neme, G. A., & Gil, A. F. (2022). Past maize consumption correlates with population change in central western Argentina. *Journal of Anthropological Archaeology*, 68, 101457.